

sync-misc

read-copy-update (high-level overview)

idea: read-mostly data structure

when reading:

- read normally *via shared pointer*

when writing:

- make a copy

- atomically update the *shared pointer*

- delete the old version *eventually*

tricky part: when is it safe to delete old version

- implementation: *scheduler integration*

RCU operations

read lock — record: “I am reading now”

read unlock — record: “I am done reading now”

publish — atomically update pointer

after publish: wait until

- all threads currently running have context switched

- ... and none of them set the “I am reading now” bit

backup slides